



NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Third Semester B. Tech (CBCS) Degree Examinations
 Summer Semester July 2024

CS2001-1 – DATA STRUCTURES
 (CS, IS, AI&ML, CCE, AI&DS)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.

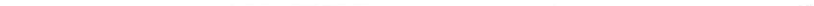
PART - A: MULTIPLE CHOICE QUESTIONS**20 Marks**

- Which of the following is a collection of different data types?
 A) Array
 B) Structure
 C) Pointer
 D) char
- What will be the output of the following C code?

```
#include <stdio.h>
int main()
{
    int *ptr, a = 10;
    ptr = &a;
    *ptr += 1;
    printf("%d,%d\n", *ptr, a);
}
```

 A) 10,10
 B) 10,11
 C) 11,10
 D) 11,11
- Presence of code like `s.b = 10` indicates
 A) Syntax Error
 B) Structure
 C) double data type
 D) An ordinary variable name
- If the elements "B", "A", "D" and "C" are placed in a stack and are deleted one at a time, what is the order of removal?
 A) DCBA
 B) ABCD
 C) CDAB
 D) CABD
- Which of the following is essential for converting an infix expression to the postfix form efficiently?
 A) An operand stack
 B) An operator stack
 C) An operand stack and an operator stack
 D) An integer stack
- Which of the following is not an inherent application of stack?
 A) Job scheduling
 B) Evaluation of postfix expression
 C) Implementation of recursion
 D) Reversing a string
- Which of the following statements is true?
 A) Recursion is always better than iteration
 B) Recursion uses more memory compared to iteration
 C) Recursion uses less memory compared to iteration
 D) Iteration is always better and simpler than recursion
- The five items P, Q, R, S and T are pushed in a stack, one after the other starting from P. The stack is popped four times and each element is inserted in a queue. Then two elements are deleted from the queue and pushed back on the stack. Now one item is popped from the stack. The popped item is:
 A) P
 B) Q
 C) S
 D) R

- ```
void solve(struct node* start)
{
 if(start == NULL)
 return;
 printf("%d ", start->data);
 if(start->next != NULL)
 solve(start->next->next);
 printf("%d ", start->data);
}
```

12. 

```
struct node *P
P=Frist ->link->link
P->link->link->link=Frist
P=P->link->link
printf("%c", P->data);
```

18. The number of external nodes in a full binary tree with  $n$  internal nodes is?

A)  $n$                                       B)  $n+1$

C)  $2n$                                      D)  $2n+1$

19. What is a threaded binary tree traversal?  
 A) a binary tree traversal using stacks and queues  
 B) a binary tree traversal using stacks  
 C) a binary tree traversal without using stacks and queues  
 D) a binary tree traversal using queues
20. What is the data Structure used in hash tables?  
 A) Stack  
 B) Array  
 C) Tree  
 D) Queue

**PART - B: DESCRIPTIVE ANSWER QUESTIONS****Unit – I**

|                                                                                                                                                                                                                                                            | Marks | BT* | CO* | PO* |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-----|-----|-----|
| 1. a) With syntax explain how dynamic memory allocation can be made to have an array of n integer values and show how capacity can be increased to hold (n+m) elements without disturbing n elements entered previously                                    | 6     | L*3 | 1   | 2,3 |
| b) Design a C program to enter details of n students Roll_No, Name, GPA and DOB (Day,Month,Year) using nested structure concept and display the student details. Also write function to check whether any two given students details are same or different | 6     | L3  | 1   | 2,3 |
| c) With Suitable declaration explain in brief the following<br>(i) Array of pointers and pointer to array<br>(ii) Array of structures and array within structure                                                                                           | 4     | L2  | 1   | 1,3 |
| 2. a) Give a proper data structure for an integer stack, Design C functions for primitive operations of stack with proper testing conditions to avoid illegal operations. Show how a stack can be used to reverse a given number                           | 6     | L3  | 2   | 2,3 |
| b) Design an algorithm to evaluate a postfix expression and Using evaluation of a postfix expression algorithm, show the conversion of postfix expression ABC*+DE/- to infix expression                                                                    | 6     | L3  | 2   | 2,3 |
| c) Convert the following infix expressions to prefix and postfix expressions:<br>(i) (A+B*C-D) / E<br>(ii) ((A+B)*C-(D-E))\$(F+G)                                                                                                                          | 4     | L2  | 2   | 1   |
| 3. a) Design an algorithm to convert infix to postfix expression. Show the steps of conversion of the infix expression (A+B)*(C-D)/E to postfix expression using that convert algorithm                                                                    | 6     | L3  | 2   | 2,3 |
| b) Define Tower of Hanoi problem. Design a recursive algorithm to solve tower of Hanoi problem and show tracing it for number of disk = 3                                                                                                                  | 6     | L3  | 2   | 2,3 |
| c) Give the recursive definitions and algorithms to compute<br>(i) n <sup>th</sup> Fibonacci<br>(ii) GCD(m,n)                                                                                                                                              | 4     | L2  | 2   | 1,2 |

**Unit – II**

|                                                                                                                                                                                                                                                       |   |    |   |     |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|----|---|-----|
| 4. a) What is the limitation of ordinary Queue? Explain in brief how it is resolved in circular Queue. Design C functions for inserting and deleting an element to and from a circular queue with proper data structure to represent a circular queue | 6 | L2 | 3 | 2,3 |
| b) What is a Linked list? With a node structure and primitive operation show how can you implement queue using a singly linked list.                                                                                                                  | 6 | L2 | 4 | 2,3 |
| c) Develop C functions to<br>(i) Place a node in a sorted linked list.<br>(ii) Concatenate two singly linked lists.                                                                                                                                   | 4 | L3 | 4 | 2,3 |
| 5. a) Define a priority queue. With a suitable example show linked representation of a priority queue. Give the algorithm to insert an ITEM with priority number N and to delete the element from priority queue.                                     | 6 | L2 | 3 | 3   |

- b) What is a circular linked list? With a node structure and primitive operation show how can you implement stack using a circular linked list.

6 L3 4 2,3

- c) Develop C functions to

- (i) Concatenate 2 Circular linked lists  
(ii) Display given circular list and its length

4 L3 4 3

6. a) Develop C Functions to

- (i) Create a DLL using insert last function  
(ii) Insert a node after the given position in that DLL  
(iii) Delete a node at the given position in that DLL.

6 L3 4 2,3

- b) With proper data structure develop the C functions to represent, read and display the polynomial  $P(x,y)=6x^2y^2+3xy^2+2x^2y+xy$  using linked list

- i) Read\_Poly()  
ii) Display\_Poly()

6 L3 4 2,3

- c) With supporting diagram explain

- (i) doubly linked list with header  
(ii) Circular doubly linked list with header

4 L2 4 2,3

### Unit – III

7. a) Show array representation for the following trees with supporting diagrams and comment on that representation.

- (i) binary tree with 5 nodes  
(ii) Complete binary tree with 9 nodes  
(iii) Fully binary tree with 7 nodes

6 L3 5 2,3

- b) Explain with supporting algorithms three methods of tree traversals and give the result of the traversals for the tree shown in figure 7b.

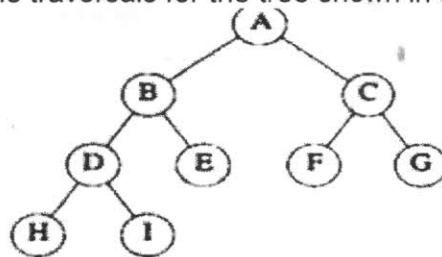


Figure 7b

6 L3 5 2,3

- c) What is a threaded binary tree? Show the pictorial representation of threaded binary tree for tree shown in figure 7b giving rules to construct that threaded tree

4 L2 5 2

8. a) Define a Binary search tree. Construct a binary search tree showing insertion of one node at each stage for the given list 30,40,35,18,20,28,80,9,20. Give a recursive algorithm to search a particular key in a binary search tree.

6 L3 5 2,3

- b) Explain the two ways of representing the given undirected graph G(6,7) giving the significance of the two methods

Graph G: Nodes: ABCDEF

Edges: AC, AD, CE, CB, EF, BF, ED

6 L3 5 2,3

- c) What is hashing? Show how the collision is resolved using chaining for the example

- Record: A, B, C, D, E, X, Y, Z  
• H(k): 4, 8, 2, 11, 4, 11, 5, 1

4 L3 5 2,3

BT\* Bloom's Taxonomy, L\* Level; CO\* Course Outcome; PO\* Program Outcome

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